

Leakage-aware CNN classification of breast ultrasound images, with external validation

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Summary

A DenseNet121 ensemble trained on BUSI reached an ensemble AUC of 0.977 on a locked internal test set, but only 0.750 on an independent Brazilian dataset (BUS-BRA). The external drop is the main finding: single-centre BUSI results should not be read as clinical performance.

The project classifies breast ultrasound lesions as benign or malignant. It first audits BUSI for near-duplicate images and builds a group-aware, leak-free split. It then compares four ImageNet-pretrained CNNs with 5-fold cross-validation, fixes a sensitivity-first threshold on training data only, and opens the test set once. Grad-CAM is checked against radiologist lesion masks, and the frozen models are tested unchanged on BUS-BRA.

A pre-registered follow-up (v2: lesion-mask auxiliary loss plus aspect-ratio crop augmentation) raised external AUC to 0.776, a paired gain of +0.027 (95% CI +0.010 to +0.044), with no significant change on the internal test. All training ran on Kaggle cloud GPUs (NVIDIA T4).

1. Data and leak-free split

About 35% of BUSI images (277 of 780) have a near-identical copy, so a random split would leak the same scan into both training and test. Every image was fingerprinted with a perceptual hash (pHash, Hamming distance ≤ 4). Near-duplicates were merged into groups with union-find, and all splits were made at group level with `StratifiedGroupKFold`.

Audit finding	Count
Images (benign / malignant / normal)	780 (437 / 210 / 133)
Near-duplicate pairs	186
Unique groups after merging	627
Pairs with conflicting labels	10
Conflicting-label groups excluded	8 groups, 18 images

Groups whose copies carry different labels (for example the same scan marked benign and malignant) were excluded, since their true label is unknown. Assertions in the split script stop the pipeline if any group crosses the train/test boundary.

The task is benign vs malignant; normal images contain no lesion and were left out. The locked test set holds 105 lesion images (70 benign, 35 malignant). The remaining images form five cross-validation folds of about 420 training and 105 validation images each.

2. Methods

All models share one training recipe, so differences come from the architecture or the single setting being tested.

- **Preprocessing:** letterbox resize to 224×224 (aspect ratio kept, black padding), grayscale repeated to 3 channels, ImageNet normalisation.
- **Augmentation (training only):** horizontal flip, rotation $\pm 10^\circ$, small shift and scale, brightness and contrast jitter. No vertical flip, because the skin surface is always at the top of the image.
- **Models:** ImageNet-pretrained ResNet50, EfficientNet-B0, DenseNet121 and ConvNeXt-Tiny (timm), fine-tuned end to end.

- **Training:** AdamW (learning rate 1e-4, weight decay 1e-4), cosine schedule, class-weighted cross-entropy, batch 32, mixed precision, up to 30 epochs, early stopping on validation AUC (patience 7), seed 42.
- **Threshold:** chosen on pooled out-of-fold predictions only, with the rule sensitivity ≥ 0.90 , which gave 0.225. Missing a cancer is treated as worse than a false alarm.
- **Test protocol:** the five fold models are frozen with that threshold before the test set is opened once. Each model votes at the threshold; the ensemble (mean probability) is reported for AUC.
- **Uncertainty:** 95% confidence intervals from 2,000 bootstrap resamples of whole duplicate groups (BUSI) or whole patients (BUS-BRA). Model comparisons use a paired bootstrap of Δ AUC.

3. Results

3.1 Architecture comparison (5-fold CV)

DenseNet121 had the best mean AUC, the best pooled out-of-fold AUC and the smallest spread, with only 7M parameters. The gaps between models (≤ 0.03 AUC) are smaller than the fold-to-fold variation, so the ranking is suggestive, not proven.

Model	Params	AUC (mean $\pm$ SD)	Pooled OOF AUC
DenseNet121	7.0M	0.933 $\pm$ 0.025	0.924
ConvNeXt-Tiny	27.8M	0.928 $\pm$ 0.031	0.903
EfficientNet-B0	4.0M	0.911 $\pm$ 0.039	0.905
ResNet50	23.5M	0.905 $\pm$ 0.039	0.902

3.2 Ablations

Removing augmentation cut AUC by 0.035 and made folds much less stable; it was the only setting with a clear effect. Class weights left AUC unchanged (-0.002). A 320×320 input ($+0.006$) and adding the 18 conflicting-label images to training ($+0.009$) gave gains within fold-to-fold noise, so the baseline recipe was kept to avoid picking a winner on the validation folds.

3.3 Locked test set (BUSI)

The pre-set target was met: test sensitivity 0.949 (95% CI 0.900–0.986) at a threshold fixed without seeing the test set.

Metric (5 fold models, threshold 0.225)	Result
AUC, mean $\pm$ SD	0.961 $\pm$ 0.019
Ensemble AUC (95% CI)	0.977 (0.947–0.997)
Sensitivity (95% CI)	0.949 (0.900–0.986)
Specificity (95% CI)	0.754 (0.665–0.835)
Missed malignant, of 35	1.8 on average (range 0–3)

Specificity varied widely between fold models (0.59–0.86) at the same threshold, which points to a calibration issue. Test AUC was above CV AUC but inside the CV fold range, so it most likely reflects an easier split rather than a better model.

3.4 Grad-CAM against lesion masks

For correctly classified test images, the hottest Grad-CAM point fell on the radiologist's lesion 82% of the time, against 14% by chance. Heatmap energy inside the lesion was $2.7 \times$ what chance predicts overall ($3.0 \times$ for benign, $2.2 \times$ for malignant lesions).

On benign images wrongly called malignant, the malignant evidence landed on the lesion no more often than chance (0.12 vs 0.13). It sat below the lesion, where posterior acoustic shadowing appears, or at image borders. No burned-in text labels were highlighted. Using the true-class target, not the malignant target, was needed to measure localisation fairly on benign images.

3.5 External validation (BUS-BRA)

Applied unchanged to BUS-BRA (1,875 images, 1,064 patients, three scanners in Brazil), ensemble AUC fell from 0.977 to 0.750.

Metric	BUSI internal test	BUS-BRA external
Images (benign / malignant)	105 (70 / 35)	1,875 (1,268 / 607)
Ensemble AUC (95% CI)	0.977 (0.947-0.997)	0.750 (0.721-0.776)
Sensitivity at 0.225 (mean of 5 models)	0.949	0.829
Specificity at 0.225 (mean of 5 models)	0.754	0.420

The drop held on every scanner (AUC 0.73-0.79), so no single device explains it. A threshold re-tuned on BUS-BRA for 90% sensitivity landed at 0.223, almost the same as before, and still gave specificity of only 0.34. The loss is in ranking benign against malignant, not in where the threshold sits.

4. v2: pre-registered mask-guided training

The v2b variant raised external AUC by 0.027 (95% CI +0.010 to +0.044) with no significant change on the internal test (Δ AUC -0.007 , 95% CI -0.027 to $+0.009$). Motivated by Grad-CAM (evidence outside the lesion) and by BUS-BRA's taller image framing, two variants were written into docs/v2_preregistration.md and committed before any v2 training:

- **v2a:** DenseNet121 plus an auxiliary lesion-mask head (1×1 convolution on the 7×7 feature map, BCE loss, weight 0.5).
- **v2b:** v2a plus aspect-ratio crop augmentation (random 50-100% width, 80-100% height, lesion always kept).

Selection used BUSI cross-validation only. Each variant's threshold was frozen from its own out-of-fold predictions, and both test sets were opened once.

	v1	v2a	v2b (selected)
BUSI CV pooled OOF AUC	0.924	0.923	0.940
BUSI test ensemble AUC	0.977	0.969	0.970
BUS-BRA ensemble AUC	0.750	0.751	0.776
Paired Δ AUC vs v1 on BUS-BRA (95% CI)	—	+0.002 (-0.016 to $+0.018$)	+0.027 ($+0.010$ to $+0.044$)
BUS-BRA sensitivity / specificity	0.83 / 0.42	0.82 / 0.39	0.70 / 0.65

The mask loss alone (v2a) did not help; the gain appeared only with the aspect-ratio crops, so framing robustness is the likely driver. v2b also made the five fold models more consistent on BUS-BRA (specificity range 0.42-0.82, against 0.16-0.63 for v1). The improvement is modest: external AUC stays far below internal AUC, and at its BUSI threshold v2b reaches only 0.70 sensitivity on BUS-BRA, short of the 0.90 target.

The aspect-crop idea came from seeing BUS-BRA images, although BUS-BRA was never used to tune a setting. The gain may therefore be smaller on other unseen hospitals.

5. Limitations and next steps

The main limitation is that training data come from one centre with about 420 images per fold, and performance on a second hospital fell sharply.

- **Small test set:** 35 malignant test cases, so each missed cancer moves test sensitivity by about 3 points.
- **Calibration:** one threshold gives very different specificity across fold models; temperature scaling or site-specific calibration is needed before any real use.
- **Residual leakage:** near-copies beyond the pHash cut-off cannot be fully ruled out, which may make the internal estimate optimistic.
- **Grad-CAM:** a coarse 7×7 map that shows correlation, not cause; the hard-case group is small (17 images).
- **v2 design:** an aspect-crop-only variant was not pre-registered, so the effects of the mask loss and the crops cannot be fully separated.

Next steps: train on several centres (BUSI plus other public sets), try domain adaptation, add calibration, and always report a held-out external dataset. The model is a research prototype, not a medical device.

6. Reproducibility, code and demo

Every experiment used fixed seeds and group-aware splits, and the five fold models reproduced identical validation AUCs across three independent runs. The published v2b weights are a rebuild of the evaluated models: four folds match exactly and fold 4 differs by 0.002 AUC.

- **Code and full results:** github.com/wajidali99/breast-ultrasound-classification
- **Live demo** (upload an image; five models vote and a Grad-CAM heatmap is shown): huggingface.co/spaces/Wajiddev99/breast-ultrasound-classifier
- **Model weights and model card:** huggingface.co/Wajiddev99/busi-densenet121-v2b
- **Compute:** Kaggle notebooks, NVIDIA T4 GPU, PyTorch with timm.

References

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